

Paper-4
BFUHS
(Radiographer)

1. **Wavelength of X-rays ranges from**
(a) 6-7 A (b) 15-20 A
(c) 0.1-0.5 A (d) 5-10 A
2. **X-rays are electrically charged**
(a) electrons (b) protons
(c) neutral (d) neutrons
3. **Radioactive substances emit all except**
(a) alpha (b) beta
(c) gamma (d) epsilon
4. **Most hazardous internally located isotopes are**
(a) alpha particles
(b) beta particles
(c) gamma particles
(d) delta particles
5. **Radiation exposure can measured by**
(a) Ionization Chamber
(b) Scintillation detector
(c) GM Counter
(d) All the above
6. **TLD badge consists of**
(a) Calcium sulphate (b) Disprosium
(c) Teflon (d) All the above
7. **In Mammography tube, anode is made of**
(a) Tungsten (b) Molybdenum
(c) Beryllium (d) Cobalt
8. **Film badge is used to measure**
(a) beta, gamma & thermal neutron radiation
(b) beta, alpha & delta radiations
(c) beta, alpha and delta radiations
(d) gamma, thermal neutrons & delta radiation
9. **Advantage of high frequency generator are**
(a) initial conversion of AC to DC waveform
(b) To ensure accurate voltage and current
(c) Excellent service and repair
(d) All the above
10. **Modalities not involving ionizing radiations are**
(a) CT Scan (b) X-rays
(c) PET (d) MRI
11. **Biological effect at the same energy level is in the order of**
(a) $\alpha > \beta > \gamma$
(b) $\alpha < \beta < \gamma$
(c) $\alpha = \beta = \gamma$
(d) None of the above
12. **Death due to radiation exposure occurs at the dose of**
(a) 100 mSv (b) >500REM
(c) 3Sv (d) 0.25-1Sv
13. **Cataract in eye with radiation exposure of 5-8Sv occurs within**
(a) one year (b) 20-30 years
(c) 5-10 years (d) 6 months
14. **Exposure can be reduced to desired level by**
(a) Providing shield
(b) by reducing duration of exposure
(c) By measuring distance between source and area under exposure
(d) All the above
15. **Minimum thickness of lead coat should be**
(a) 2mm (b) 1.5mm
(c) 0.25 mm (d) 4mm

16. Radioactivity was discovered by
(a) Roentgen (b) Becquerel
(c) Madam Curie (d) None of the above
17. Causes of unsharpness in Radiographic image are
(a) geometry (b) movement
(c) photographic (d) all the above
18. Radiographic examination of joints is called
(a) Arthrography
(b) Arteriography
(c) Dacrocystography
(d) Cholangiography
19. Photoconductor used in Xeroradiography is
(a) Cobalt (b) Tungsten
(c) Selenium (d) Copper
20. Good and uniform compression of Breast in mammography is to
(a) Immobilise breast
(b) Decrease geometric unsharpness
(c) Decrease scatter radiations
(d) All the above
21. Most commonly used lubricant purgative in abdominal preparation is
(a) Liquid paraffin (b) Isogel
(c) Bisacodyl (d) Castor oil
22. Contraindication for intravenous pyelography is
(a) Blunt injury abdomen with haematuria
(b) Pregnancy
(c) Congenital anomalies involving kidney
(d) All the above
23. Enteroclysis is
(a) Small bowel enema
(b) Barium meal follow thro
(c) T-tube cholangiography
(d) Saliography
24. Ohm's law of electric current is
(a) $V \sim I$ (b) $V \sim 1/I$
(c) $V \sim 1/R$ (d) None of the above
25. According to Maxwell's electromagnetic theory
(a) light is magnetic wave
(b) Light is electrical wave
(c) Light is transverse wave
(d) All the above
26. Quantum theory fails to explain
(a) Interference
(b) Radiation emission
(c) Photoelectric effect
(d) All the above
27. Visible spectrum ranges from
(a) 1000-3000Å (b) 4000-7700Å
(c) 8000-9900Å (d) 200-500Å
28. Contrast material used in Micturating cystourethrography
(a) Sodium Iodide (b) Barium
(c) Telepaque (d) Dulcolax
29. Investigation of choice in pericardial effusion:
(a) Cardiac catheterization
(b) USG
(c) Echocardiography
(d) Lateral view X-ray chest
30. All are imaging modalities of chest, except
(a) HRCT
(b) USG
(c) Ventilation perfusion study
(d) Echocardiography
31. SI unit for measurement of luminous intensity is
(a) Fermi (b) Parsec
(c) Candela (d) Kelvin
32. Which of the following is semi conductor
(a) Silicon (b) Germanium
(c) Carbon (d) All of the above
33. Adverse reaction to contrast are all except
(a) Convulsions
(b) Cataract
(c) Laryngeal oedema
(d) Pulmonary oedema
34. Barium suspension is made of
(a) Barium sulphate
(b) Barium carbonate
(c) Barium phosphate
(d) Barium silicate
35. Biliary contrast medium is
(a) Urograffin
(b) Barium sulphate
(c) Barium carbonate
(d) Telepaque

36. Basic elements of image intensifier are
(a) Input phosphor
(b) Accelerating anode
(c) Output phosphor
(d) All of the above
37. Walls of radiography room should be lined with
(a) Tin (b) Molybdenum
(c) Lead (d) Iron
38. Ear is located in
(a) Frontal bone (b) Temporal bone
(c) Occipital bone (d) None of the above
39. Length of ureter is
(a) 25 cm (b) 40 cm
(c) 15 cm (d) 5 cm
40. Adam's apple is due to prominence of
(a) Thyroid cartilage (b) Cricoid cartilage
(c) Arytenoid (d) Epiglottis
41. Working distance of safe light to the film should not be less than
(a) 9 metres (b) 1.2 metres
(c) 3 metres (d) None of the above
42. Wattage of bulb in safe light should be less than
(a) 100 watts (b) 50 watts
(c) 30 watts (d) 15 watts
43. Heart of X-ray film is
(a) Developer (b) Substratum
(c) Base (d) Emulsion
44. Automatic film processors have all advantages except
(a) Shortened processing time
(b) Improves quality
(c) Increases capacity of radiology dept
(d) Temperature regulation is variable
45. All are components of fixer except
(a) Hydroquinone (b) Sodium thiosulphate
(c) Sodium sulphite (d) Water
46. Use of hyoscine N Butylbromide (Buscopan) is contraindicated in
(a) Pain abdomen
(b) Bowel spasm.
(c) For adequate distension of bowel
(d) Severe Prostatism
47. Radiographic contrast depends on
(a) Type of film
(b) Intensifying screens
(c) Scatter radiations
(d) All the above
48. Penetration power of radiograph is increased by
(a) Decreasing kilo voltage
(b) Increasing FFD
(c) Increasing mAS
(d) Increasing kilo voltage
49. All are advantages of spiral CT except
(a) No motion artifact
(b) Improves lesion detection
(c) Multiplanar image
(d) Increase image noise
50. Heart of MRI system is
(a) Shim coils (b) Computer
(c) Gradient coils (d) Magnet
51. Father of Computed Tomography is
(a) Curie (b) W C Roentgen
(c) Robert Koch (d) Godfrey Hounsfield
52. Partial voluming artifact reduced by using
(a) Small voxel (b) Large voxel
(c) None (d) Both a & b
53. RF pulse causes longitudinal magnetization to _____ and establishes a new transversal magnetization
(a) Increase (b) Remain unchanged
(c) Decrease (d) Become zero
54. $1/T_2$ is also called _____
(a) Longitudinal relaxation rate
(b) Transverse relaxation rate
(c) Longitudinal frequency
(d) Transverse frequency
55. TR less than _____ is short.
(a) 300msec (b) 100msec
(c) 700msec (d) 500msec
56. T_2 of fat is _____ that of water
(a) Longer than
(b) Shorter than
(c) Equal to
(d) Inversely proportional to

57. TR more than _____ is long.
 (a) 700 msec (b) 1000 msec
 (c) 1500 msec (d) 500msec
58. We get T1 weighted image, when TR is _____
 (a) Short
 (b) Long
 (c) Too to be measured
 (d) None of the above
59. In a normal case, air is seen as dark on
 (a) CT image
 (b) T1weighted image
 (c) T2 weighted image
 (d) All the above
60. In a normal case, dense bone is seen as dark on
 (a) CT image
 (b) MR image
 (c) Both MR and CT images
 (d) None of CT or MR images
61. In medical Ultrasonography, Piezoelectric material used is
 (a) Calcium Tungstate
 (b) Zinc cadmium sulphate
 (c) Lead zirconate titanate.
 (d) None of the above.
62. To sterilise the USG transducer, it should be
 (a) Autoclaved.
 (b) Heated.
 (c) Cleaned with spirit.
 (d) None of the above.
63. USG produces which of the following biological effects.
 (a) Heat (b) Cavitation
 (c) None of these. (d) Both of these.
64. In Color Doppler venous flow is normally which type?
 (a) Plug flow (b) Circular flow
 (c) Laminar flow (d) None of the above
65. Unit of Pulse repetition frequency is
 (a) Per minute (b) msec
 (c) mm/sec (d) MHz
66. In which of the following, Doppler has two piezoelectric elements with single head?
 (a) CW Doppler
 (b) Pulse Doppler
 (c) Duplex
 (d) Doppler color flow imaging
67. Transrectal scanner is best at which frequency?
 (a) 3-7.5 (b) 5-7.5
 (c) 3-5 (d) 8-12
68. Contrast resolution in CT is
 (a) Same as conventional films
 (b) Less than conventional films
 (c) Better than conventional films
 (d) None of the above
69. CT number of water is
 (a) zero (b) -1000
 (c) 200-300 (d) +1000
70. CT number of bone is
 (a) zero (b) +1000
 (c) 200-300 (d) -1000
71. In CT reference material is
 (a) water (b) air
 (c) bone (d) none of the above
72. With narrowing of the window, each gray scale covers
 (a) less CT numbers
 (b) larger CT numbers
 (c) no effect on CT numbers
 (d) none of the above
73. Barium swallow is indicated in all except
 (a) Motility disorder
 (b) Tracheo Esophageal fistulae
 (c) Failed upper GI endoscopy
 (d) Intestinal obstruction
74. Virtual Colonoscopy (CT colonoscopy) is the radiological investigation of choice for
 (a) Detection of Lung cancer
 (b) Acid Peptic disease.
 (c) Detection of Colonic neoplasia.
 (d) Detection of esophageal stricture.
75. MR imaging of GI tract is indicated in
 (a) Perianal fistula
 (b) Local staging of anorectal cancer
 (c) Inflammatory bowel disease.
 (d) All the above.

76. **Indication for MCU is**
(a) Vesicoureteric reflux
(b) To study urethral pathologies
(c) Bladder leak.
(d) All the above.
77. **CT scan of urinary tract is done**
(a) Renal colic/ renal stone disease
(b) Intestinal mass
(c) Gall stones
(d) Acute Pancreatitis
78. **Indication for MR Urography**
(a) Congenital anomalies
(b) To determine the level of obstruction.
(c) Urinary obstruction unrelated to urolithiasis
(d) all the above
79. **IVU is contraindicated in**
(a) Haematuria
(b) Recurrent urinary tract
(c) Acute renal colic
(d) Vesical calculus.
80. **To evaluate Gall Bladder on imaging, patient should be fasting for**
(a) 6hrs (b) 24 hrs
(c) 2 hrs (d) 18 hrs
81. **ERCP is contraindicated in**
(a) Post Cholecystectomy syndrome
(b) Management of bile duct stones.
(c) Severe cardiac disease
(d) Treatment of Chronic pancreatitis.
82. **In MR, if fluid is darker than solids, the image is _____.**
(a) PD weighted image
(b) T1 weighted image
(c) T2 weighted image
(d) None of the above.
83. **If fluid e.g. CSF or urine is white, image is _____.**
(a) PD weighted image
(b) T1 weighted image
(c) T2 weighted image
(d) None of the above.
84. **In dual source CT, which of the following filters are used?**
(a) Lead (b) Tungsten
(c) Tin (d) Rhenium
85. **The Ultrasound frequency used in the diagnosis is**
(a) 20 KHz (b) 40 KHz
(c) 1MHz - 20MHz (d) 60kHz.
86. **On applying electric field certain materials change**
(a) Chemical dimension
(b) Physical dimension
(c) Both a&b.
(d) None of the above.
87. **Which of the following Phosphor is not used in intensifying screens**
(a) Calcium tungstate.
(b) Zinc cadmium sulphide
(c) Terbium
(d) Thulium blue
88. **In CT scan the 4th generation is**
(a) Translate rotate type
(b) Rotate- rotate type
(c) Stationary - rotate type
(d) None of the above.
89. **5th generation CT scan uses**
(a) Pencil beam of x-rays
(b) Narrow fan beam.
(c) Broad fan beam
(d) Multiple x-ray beams.
90. **Which of the following is best lateral resolution?**
(a) 15cm (b) 6cm
(c) 0.06cm (d) 2cm.
91. **Which of the following statements regarding protons are correct?**
(a) They have a negative charge
(b) They are equal to the number of electrons in a non-ionized atom
(c) They have no mass
(d) Hydrogen atom has 2 protons
92. **Which of the following is not correct for Tungsten:**
(a) It has an atomic number of 74
(b) The K-shell binding energy of tungsten is 69.5 KeV
(c) The mass number of tungsten is 284
(d) It is represented by the letter W

93. Which of following is correct for electromagnetic radiation:
- It cannot travel in vacuum.
 - Travels with a speed of light in vacuum.
 - Visible light is not a part of electromagnetic spectrum
 - X-rays and gamma rays have different frequency and wavelength
94. In a diagnostic X-ray tube:
- Anode is positive
 - Cathode is made of aluminium
 - Cathode is positive
 - X-rays are produced by thermionic emission
95. Cathode of X-ray tube is:
- Is commonly made of tungsten
 - Has a low melting point
 - Has a low resistance
 - Is positively charged in relation to the anode
96. The following techniques can be used to minimise scatter
- Using collimation
 - Air gaps between the object and the image receptor
 - Intensifying screens
 - Increasing the tube kV
97. Which of the following is correct for mammography tube:
- Tube voltage of 40-50 kVp
 - Molybdenum target anode
 - Single-phase voltage supply
 - Molybdenum window
98. All of these are deterministic effects of radiation except-
- Sterility
 - Leukaemia
 - Formation of cataract
 - Hair loss
99. Regarding computed tomography (CT) imaging which of these is true-
- CT number of fat is higher than water.
 - High level of scatter reaches the CT detectors.
 - Windowing is a technique to adjust the greyscale.
 - Air corresponds to a CT value of 0 HU
100. All of the following are true for X-rays except
- They are invisible
 - They form part of electromagnetic spectrum
 - They cannot travel through vacuum
 - They cause ionization

Answer and Explanation

1. Ans. (c) 0.1-0.5 Å

Exp. The wavelength of X-rays ranges from 0.1 to 0.5 Angstroms.

2. Ans. (c) neutral

Exp. X-rays are electrically neutral.

3. Ans. (d) epsilon

Exp. Radioactive substances emit alpha, beta, and gamma radiation, but not epsilon radiation.

4. Ans. (a) alpha particles

Exp. Alpha particles are the most hazardous when internally located due to their high ionizing power.

5. Ans. (d) All the above

Exp. Radiation exposure can be measured by Ionization Chamber, Scintillation detector, and GM Counter.

6. Ans. (d) All the above

Exp. A TLD badge consists of Calcium sulphate, Dysprosium, and Teflon.

7. Ans. (b) Molybdenum

Exp. In a mammography tube, the anode is made of molybdenum to produce X-rays suitable for imaging breast tissue.

8. Ans. (a) beta, gamma & thermal neutron radiation

Exp. A film badge is used to measure beta, gamma, and thermal neutron radiation.

9. Ans. (d) All the above

Exp. Advantages of a high-frequency generator include the initial conversion of AC to DC waveform, ensuring accurate voltage and current, and excellent service and repair.

10. Ans. (d) MRI

Exp. MRI (Magnetic Resonance Imaging) does not involve ionizing radiation.

11. Ans. (a) alpha > beta > gamma

Exp. The biological effect at the same energy level is in the order of alpha > beta > gamma, with alpha particles being the most biologically damaging.

12. Ans. (b) >500REM

Exp. Death due to radiation exposure occurs at doses greater than 500 REM.

13. **Ans. (a) one year**

Exp. Cataract formation in the eye with radiation exposure of 5-8 Sv occurs within one year.

14. **Ans. (d) All the above**

Exp. Exposure can be reduced to a desired level by providing a shield, reducing the duration of exposure, and measuring the distance between the source and the area under exposure.

15. **Ans. (c) 0.25 mm**

Exp. The minimum thickness of a lead coat should be 0.25 mm.

16. **Ans. (b) Becquerel**

Exp. Radioactivity was discovered by Henri Becquerel.

17. **Ans. (d) all the above**

Exp. Causes of unsharpness in a radiographic image include geometry, movement, and photographic factors.

18. **Ans. (a) Arthrography**

Exp. The radiographic examination of joints is called arthrography.

19. **Ans. (c) Selenium**

Exp. The photoconductor used in xeroradiography is selenium.

20. **Ans. (d) All the above**

Exp. Good and uniform compression of the breast in mammography is to immobilize the breast, decrease geometric unsharpness, and decrease scatter radiations.

21. **Ans. (d) Castor oil**

Exp. The most commonly used lubricant purgative in abdominal preparation is castor oil.

22. **Ans. (b) Pregnancy**

Exp. Intravenous pyelography is contraindicated in pregnancy.

23. **Ans. (a) Small bowel enema**

Exp. Enteroclysis is a small bowel enema.

24. **Ans. (d) None of the above**

25. **Ans. (d) All the above**

Exp. According to Maxwell's electromagnetic theory, light is a magnetic wave, an electrical wave, and a transverse wave.

26. **Ans. (a) Interference**

Exp. Quantum theory fails to explain interference.

27. **Ans. (b) 4000-7700 Å**

Exp. The visible spectrum ranges from 4000 to 7700 Angstroms.

28. **Ans. (a) Sodium Iodide**

Exp. The contrast material used in micturating cystourethrography is sodium iodide.

29. **Ans. (c) Echocardiography**

Exp. The investigation of choice in pericardial effusion is echocardiography.

30. **Ans. (b) USG**

Exp. USG (ultrasound) is not an imaging modality of the chest for detailed structural imaging.

31. **Ans. (c) Candela**

Exp. The SI unit for the measurement of luminous intensity is the candela.

32. **Ans. (d) All of the above**

Exp. Silicon, Germanium, and Carbon are semiconductors.

33. **Ans. (b) Cataract**

Exp. Adverse reactions to contrast do not include cataract formation.

34. **Ans. (a) Barium sulphate**

Exp. Barium suspension is made of barium sulphate.

35. **Ans. (a) Urografin**

Exp. Urografin is a biliary contrast medium.

36. **Ans. (d) All of the above**

Exp. The basic elements of an image intensifier are the input phosphor, accelerating anode, and output phosphor.

37. **Ans. (c) Lead**

Exp. The walls of a radiography room should be lined with lead for radiation protection.

38. **Ans. (b) Temporal bone**

Exp. The ear is located in the temporal bone.

39. **Ans. (a) 25 cm**

Exp. The length of the ureter is approximately 25 cm.

40. **Ans. (a) Thyroid cartilage**

Exp. The Adam's apple is due to the prominence of the thyroid cartilage.

41. **Ans. (b) 1.2 metres**

Exp. The working distance of a safelight to the film should not be less than 1.2 metres.

42. **Ans. (d) 15 watts**

Exp. The wattage of a bulb in a safelight should be less than 15 watts.

43. **Ans. (d) Emulsion**

Exp. The heart of an X-ray film is the emulsion.

44. **Ans. (d) Temperature regulation is variable**

- Exp.** Automatic film processors have all advantages except that temperature regulation is variable.
- 45. Ans. (a) Hydroquinone**
- Exp.** Hydroquinone is not a component of the fixer solution.
- 46. Ans. (d) Severe Prostatism**
- Exp.** The use of hyoscine N Butylbromide (Buscopan) is contraindicated in severe prostatism.
- 47. Ans. (d) All the above**
- Exp.** Radiographic contrast depends on the type of film, intensifying screens, and scatter radiations.
- 48. Ans. (d) Increasing kilo voltage**
- Exp.** The penetration power of a radiograph is increased by increasing the kilo voltage (kV).
- 49. Ans. (d) Increase image noise**
- Exp.** All are advantages of spiral CT except that it increases image noise.
- 50. Ans. (d) Magnet**
- Exp.** The heart of an MRI system is the magnet.
- 51. Ans. (d) Godfrey Hounsfield**
- Exp.** The father of Computed Tomography (CT) is Godfrey Hounsfield.
- 52. Ans. (a) Small voxel**
- Exp.** Partial voluming artifact is reduced by using a small voxel.
- 53. Ans. (c) Decrease**
- Exp.** RF pulse causes longitudinal magnetization to decrease and establishes a new transversal magnetization.
- 54. Ans. (b) Transverse relaxation rate**
- Exp.** $1/T_2$ is also called the transverse relaxation rate.
- 55. Ans. (a) 300msec**
- Exp.** TR less than 300 msec is considered short.
- 56. Ans. (b) Shorter than**
- Exp.** The T_2 of fat is shorter than that of water.
- 57. Ans. (b) 1000 msec**
- Exp.** TR more than 1000 msec is considered long.
- 58. Ans. (a) Short**
- Exp.** We get a T_1 weighted image when the TR is short.
- 59. Ans. (d) All the above**
- Exp.** In a normal case, air is seen as dark on CT images, T_1 -weighted images, and T_2 -weighted images.
- 60. Ans. (c) Both MR and CT images**
- Exp.** In a normal case, dense bone is seen as dark on both MR and CT images.
- 61. Ans. (c) Lead zirconate titanate**
- Exp.** In medical ultrasonography, the piezoelectric material used is lead zirconate titanate.
- 62. Ans. (c) Cleaned with spirit**
- Exp.** To sterilize the USG transducer, it should be cleaned with spirit.
- 63. Ans. (d) Both of these**
- Exp.** USG produces biological effects such as heat and cavitation.
- 64. Ans. (c) Laminar flow**
- Exp.** In color Doppler, venous flow is normally of the laminar flow type.
- 65. Ans. (d) MHz**
- Exp.** The unit of pulse repetition frequency is MHz (megahertz).
- 66. Ans. (a) CW Doppler**
- Exp.** In CW (Continuous Wave) Doppler, there are two piezoelectric elements with a single head.
- 67. Ans. (b) 5-7.5 MHz**
- Exp.** The transrectal scanner is best at the frequency range of 5-7.5 MHz.
- 68. Ans. (c) Better than conventional films**
- Exp.** The contrast resolution in CT is better than conventional films.
- 69. Ans. (a) zero**
- Exp.** The CT number of water is zero.
- 70. Ans. (b) +1000**
- Exp.** The CT number of bone is +1000.
- 71. Ans. (a) water**
- Exp.** In CT, the reference material is water.
- 72. Ans. (a) less CT numbers**
- Exp.** With narrowing of the window, each gray scale covers fewer CT numbers.
- 73. Ans. (d) Intestinal obstruction**
- Exp.** Barium swallow is indicated in all except intestinal obstruction.
- 74. Ans. (c) Detection of Colonic neoplasia**
- Exp.** Virtual colonoscopy (CT colonoscopy) is the radiological investigation of choice for the detection of colonic neoplasia.
- 75. Ans. (d) All the above**
- Exp.** MR imaging of the GI tract is indicated in perianal fistula, local staging of anorectal cancer, and

inflammatory bowel disease.

76. **Ans. (d) All the above**

Exp. Indications for MCU include vesicoureteric reflux, studying urethral pathologies, and bladder leak.

77. **Ans. (a) Renal colic/renal stone disease**

Exp. CT scan of the urinary tract is done for renal colic/renal stone disease.

78. **Ans. (d) All the above**

Exp. Indications for MR urography include congenital anomalies, determining the level of obstruction, and urinary obstruction unrelated to urolithiasis.

79. **Ans. (c) Acute renal colic**

Exp. IVU is contraindicated in acute renal colic.

80. **Ans. (a) 6 hrs**

Exp. To evaluate the gall bladder on imaging, the patient should be fasting for 6 hours.

81. **Ans. (c) Severe cardiac disease**

Exp. ERCP is contraindicated in severe cardiac disease.

82. **Ans. (b) T1 weighted image**

Exp. In MR, if fluid is darker than solids, the image is a T1 weighted image.

83. **Ans. (c) T2 weighted image**

Exp. If fluid such as CSF or urine is white, the image is a T2 weighted image.

84. **Ans. (c) Tin**

Exp. In dual-source CT, the filters used are tin filters.

85. **Ans. (c) 1MHz - 20MHz**

Exp. The ultrasound frequency used in diagnosis is 1MHz - 20MHz.

86. **Ans. (b) Physical dimension**

Exp. On applying an electric field, certain materials change their physical dimension.

87. **Ans. (d) Thulium blue**

Exp. Thulium blue is not used in intensifying screens.

88. **Ans. (c) Stationary-rotate type**

Exp. The 4th generation CT scan is a stationary-rotate type.

89. **Ans. (c) Broad fan beam**

Exp. The 5th generation CT scan uses a broad fan beam.

90. **Ans. (c) 0.06 cm**

Exp. The best lateral resolution is 0.06 cm.

91. **Ans. (b) They are equal to the number of electrons in a non-ionized atom**

Exp. Protons are equal to the number of electrons in a non-ionized atom.

92. **Ans. (c) The mass number of tungsten is 284**

Exp. The mass number of tungsten is not 284; it is 184.

93. **Ans. (b) Travels with a speed of light in vacuum**

Exp. Electromagnetic radiation travels with the speed of light in a vacuum.

94. **Ans. (a) Anode is positive**

Exp. In a diagnostic X-ray tube, the anode is positive.

95. **Ans. (a) Is commonly made of tungsten**

Exp. The cathode of an X-ray tube is commonly made of tungsten.

96. **Ans. (b) Air gaps between the object and the image receptor**

97. **Ans. (b) Molybdenum target anode**

Exp. A mammography tube has a molybdenum target anode.

98. **Ans. (b) Leukaemia**

Exp. Leukaemia is not a deterministic effect of radiation; it is a stochastic effect.

99. **Ans. (c) Windowing is a technique to adjust the greyscale**

Exp. Windowing is a technique to adjust the greyscale in CT imaging.

100. **Ans. (c) They cannot travel through a vacuum**

Exp. X-rays can travel through a vacuum; this statement is incorrect.

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